Low-cost and easy experiments about the water in the atmosphere.

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Introduction

- Atmospheric water and the changes it suffers are essential in the water cycle.
- The concepts related to water are present in the educational curricula of many levels.
- There are many misconceptions from students related with atmospheric water.
- We present some easy experiments which can help in the learning of these concepts.

Why low-cost?

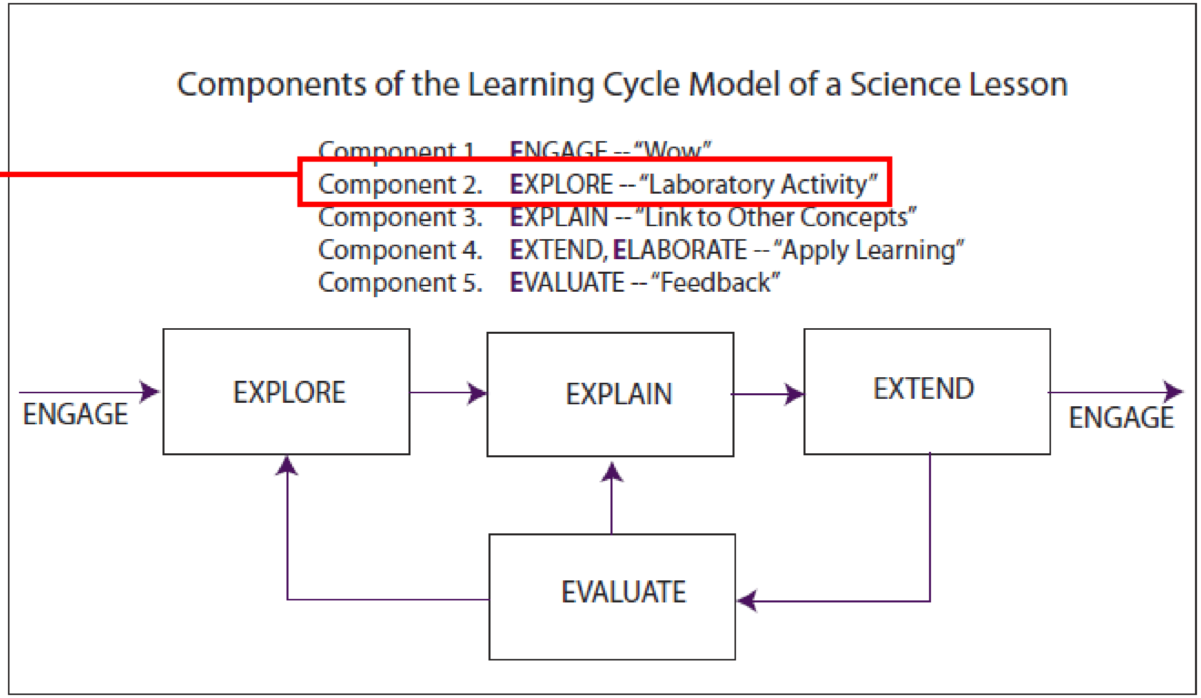
- Cheap and simple to do: promotes students autonomy.
- It only requires part of the time of a class session in the laboratory or in the classroom.
- It can be adapted to many different levels.

Ocean of misconceptions

Some examples:

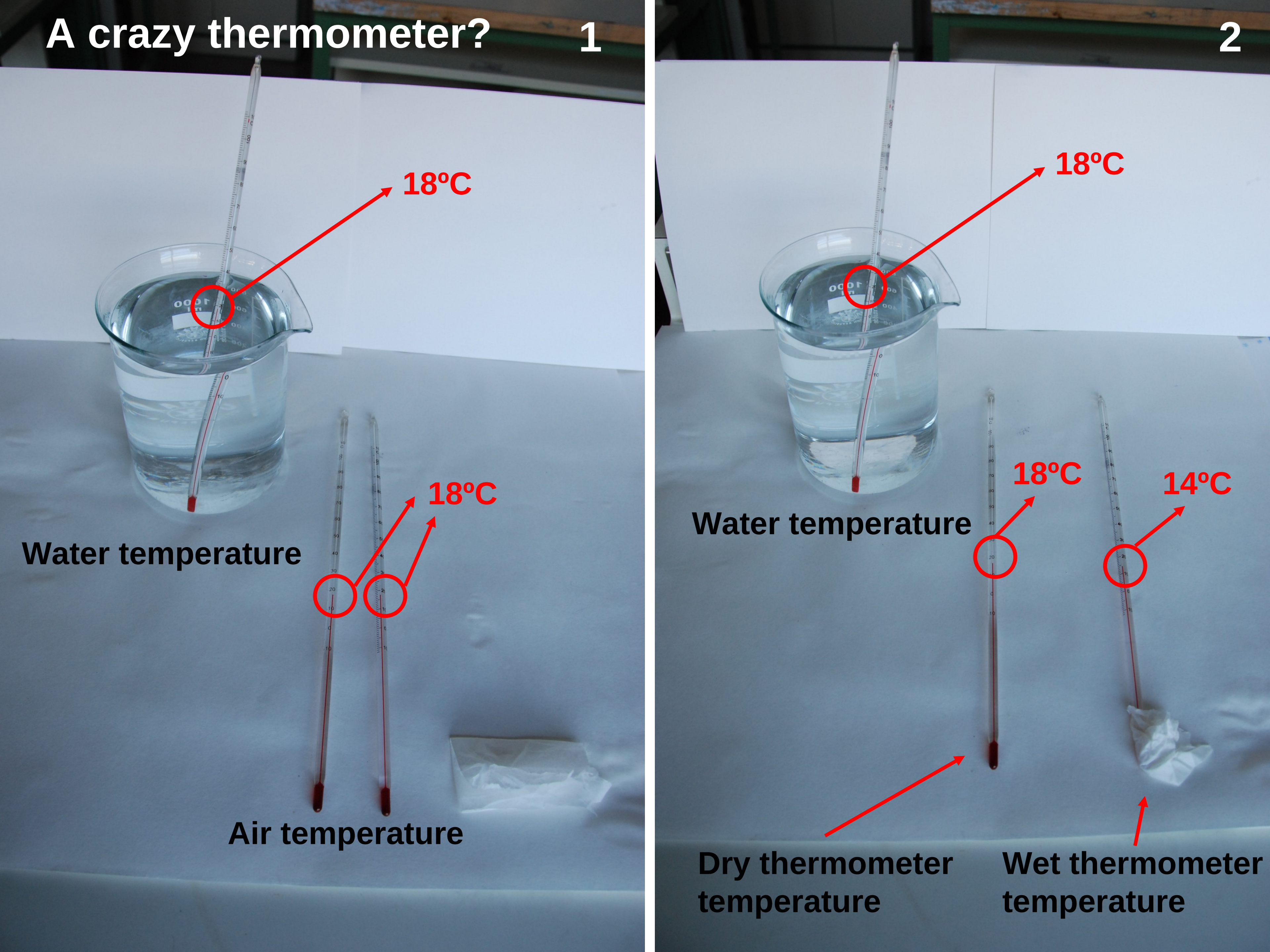
- **Clouds are made of water vapor.**
- **Air humidity and temperature are independent parameters.**
- **In cloud formation, only the water vapor is needed.**

Didactics



Teaching POE sequence

Phase	Activity
Before the experiment: **P**REDICT	What will happen? Why will it happen?
During the experiment: **O**BSERVE	Do the experiment
After the experiment: **E**XPLAIN	What really happened? Why did it happen?


A crazy thermometer?
1
18ºC
18ºC
Water temperature
Air temperature
2
18ºC
18ºC
14ºC
Water temperature
Dry thermometer
temperature
Wet thermometer
temperature

A crazy thermometer?

Key concepts: evaporation, vaporization heat, air humidity.

Key questions:

What's the initial temperature of all thermometers?

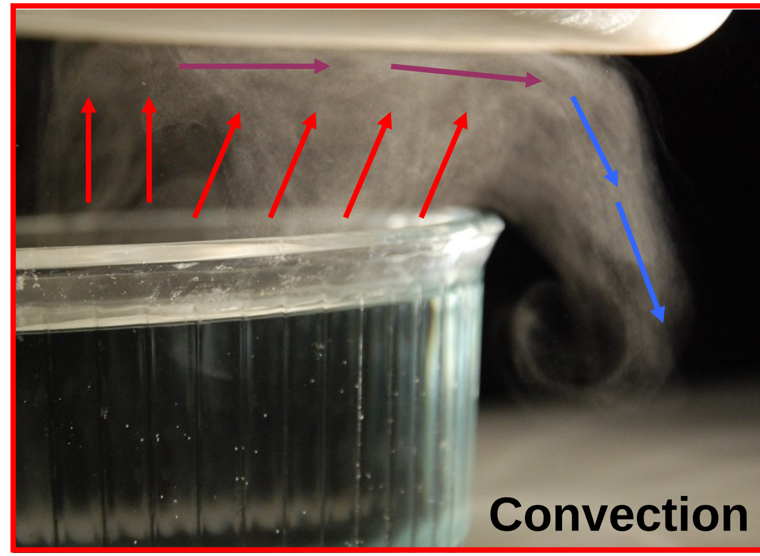
Which thermometer is cooler? Why?

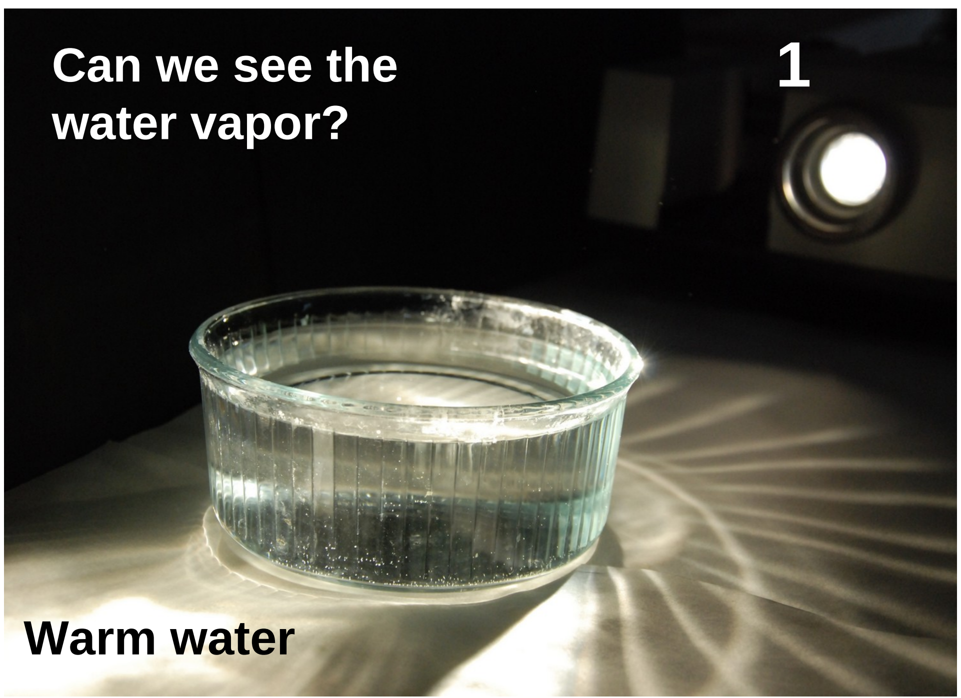
Daily facts

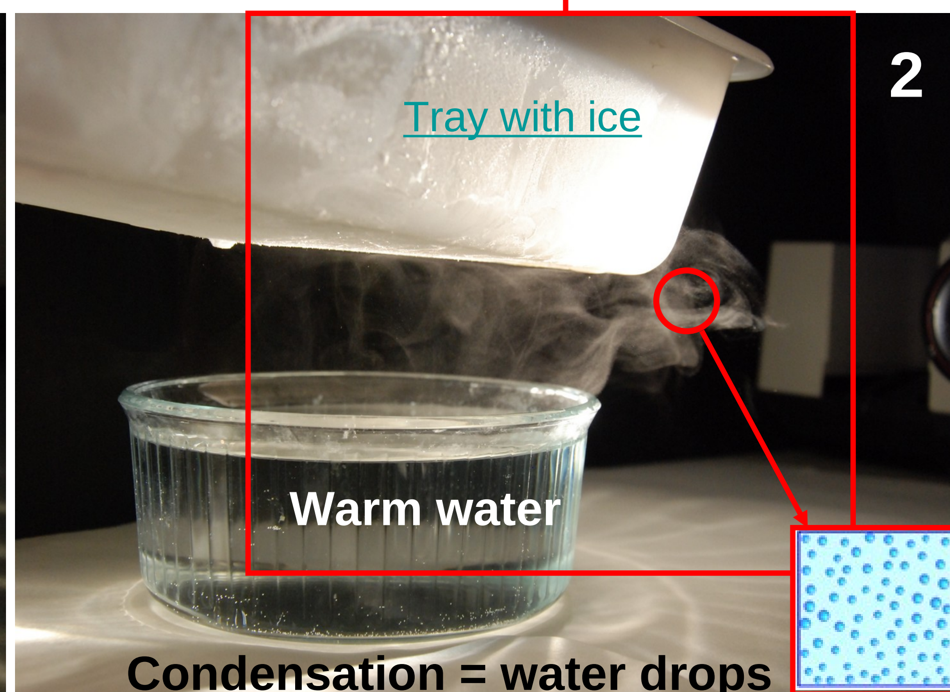
Drying clothes

Biological mechanisms of cooling

What are clouds made of?







What are clouds made of?

Key concepts: evaporation, air humidity and cooling, condensation

Key questions: Is the water in the vessel evaporating?
Is water vapor visible?
What happens when air cools?
What can we see?

Daily facts

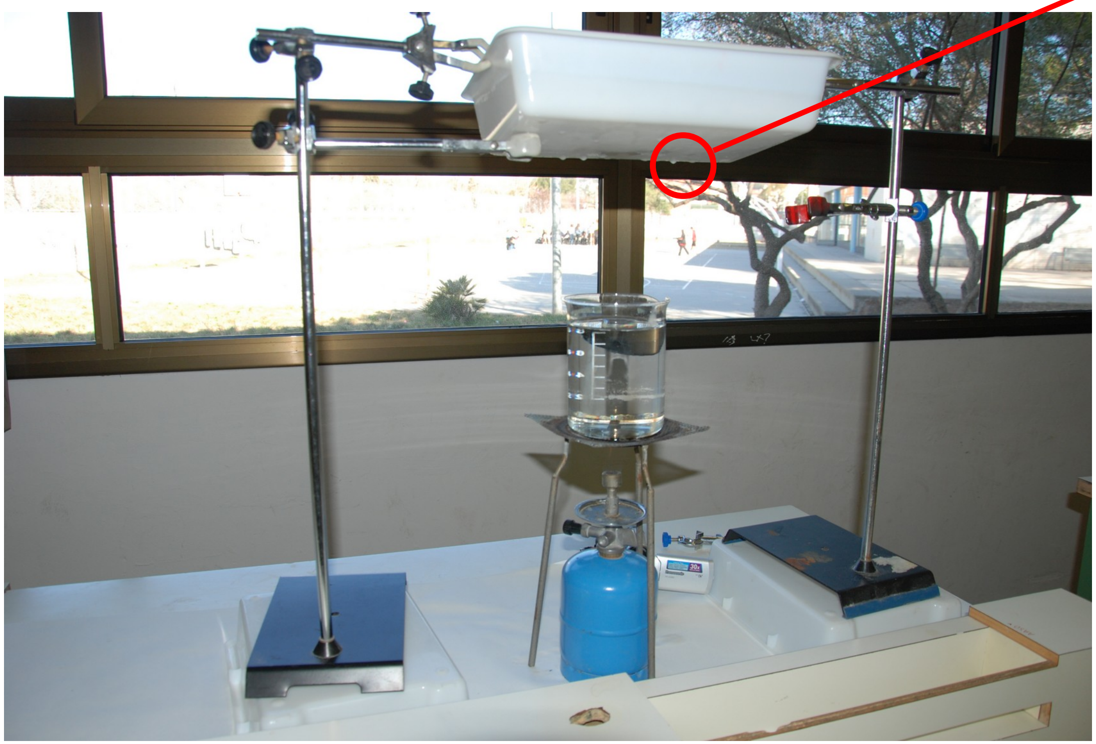
Skies and clouds

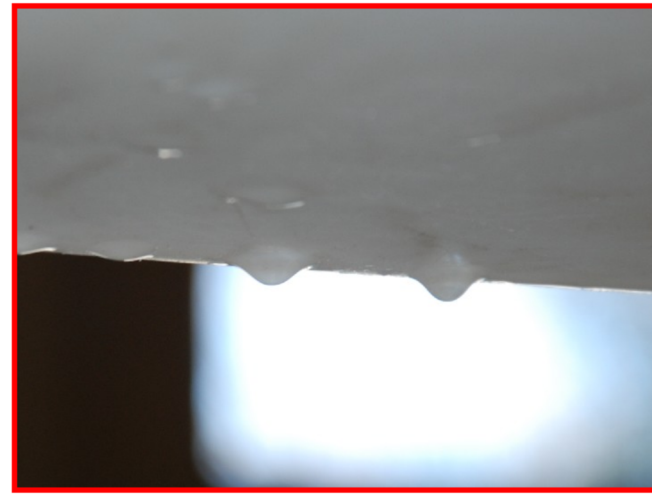
Cooking

Moisture manifestations

Does the water follow a cycle?

What material do we need?





It's going to rain: Our water cycle works!

Does the water follow a cycle?

Key concepts: evaporation, condensation, precipitation, air cooling.

Key questions:

Why does water evaporate?

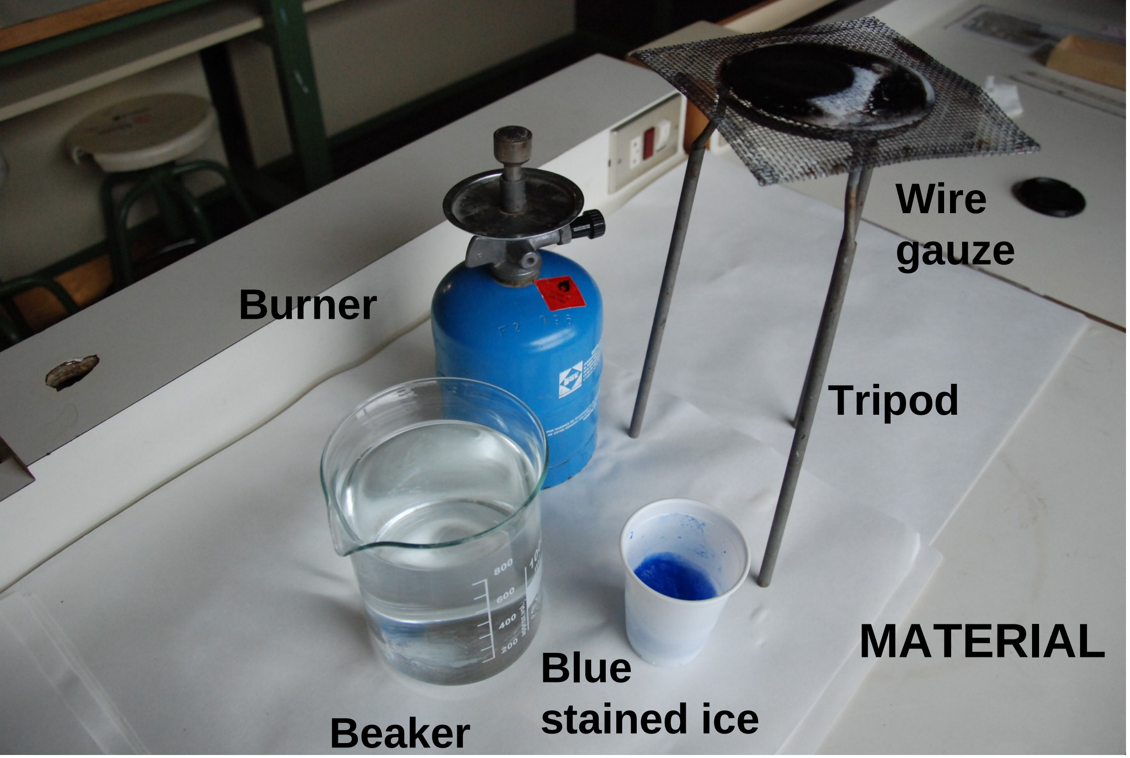
Why do appear drops in the tray?

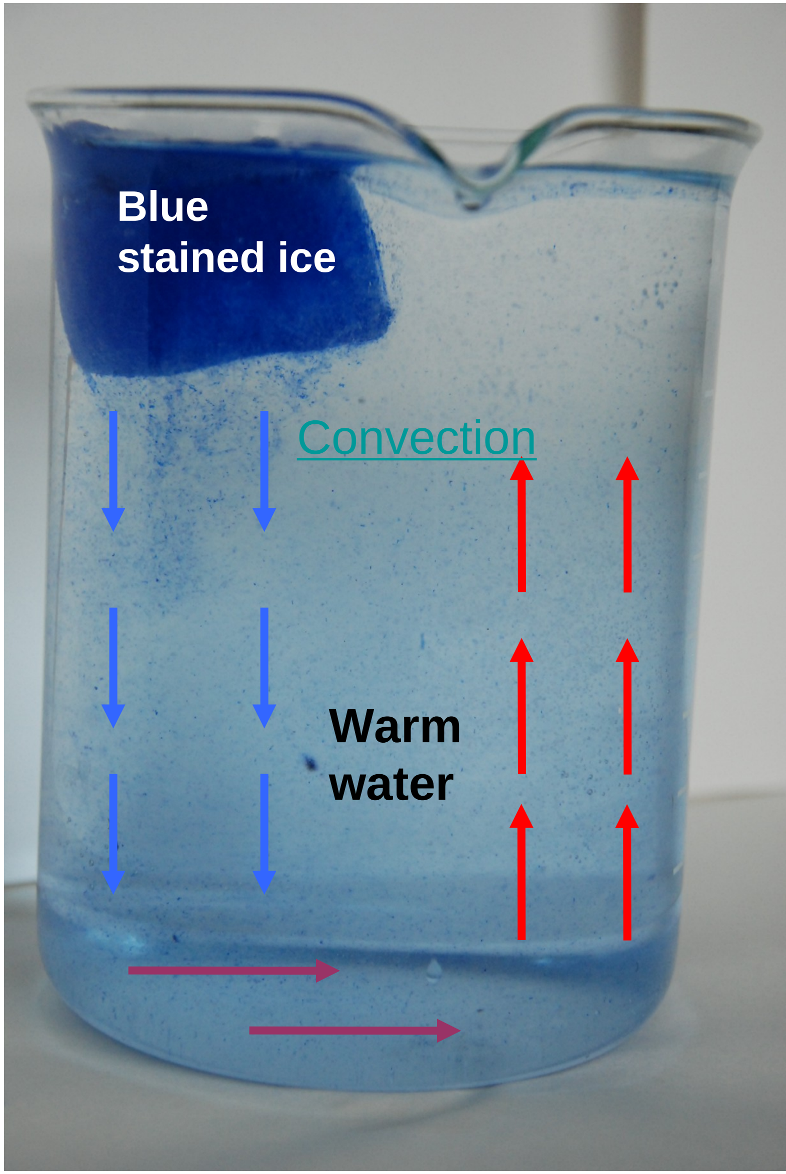
Why do drops grow and fall?

Daily facts

Hydrometeors

Water cycle elements


Wire
gauze
Burner
Tripod
MATERIAL
Blue
stained ice
Beaker


Blue
stained ice
Convection
Warm
water

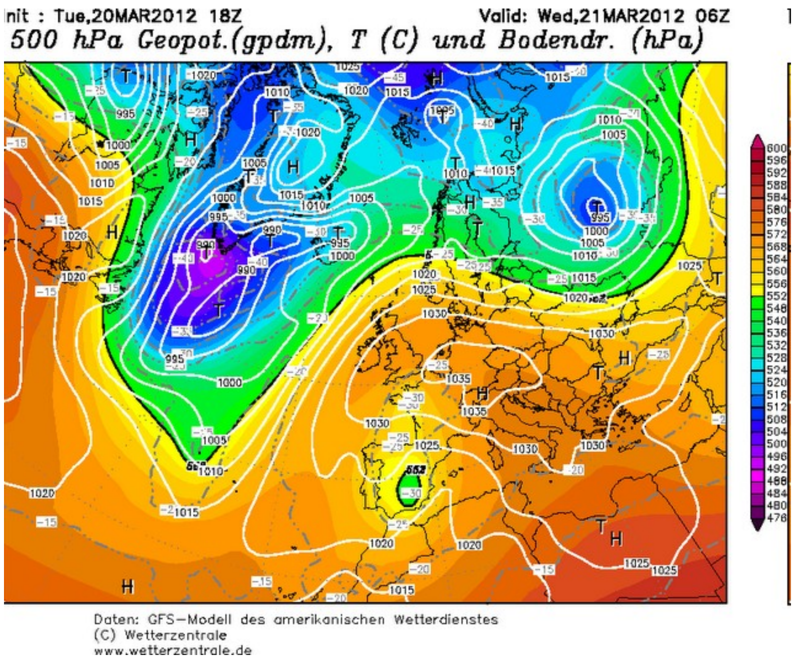
The cold drop

<u>Key concepts</u>: convection, temperature contrast, air cooling, condensation.

<u>Key questions</u>:
How are the particles in the water moving?

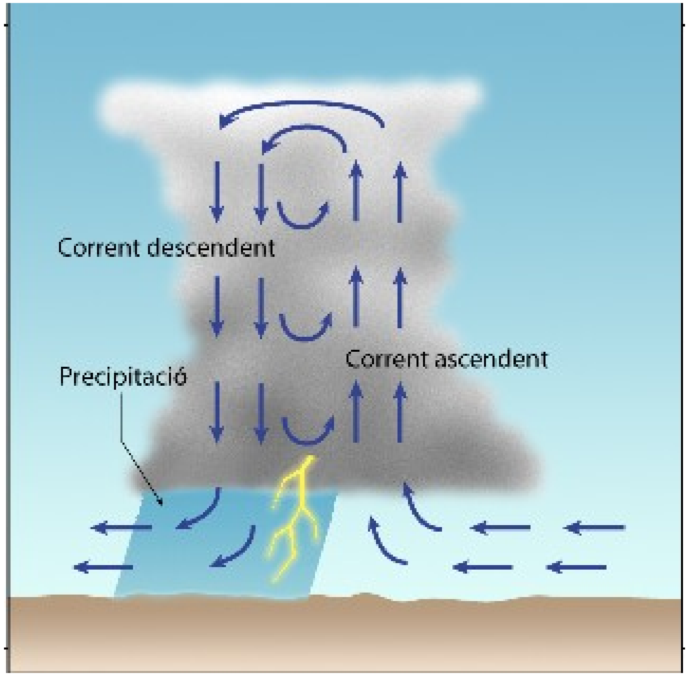
What atmospheric phenomena could originate a movement like this?

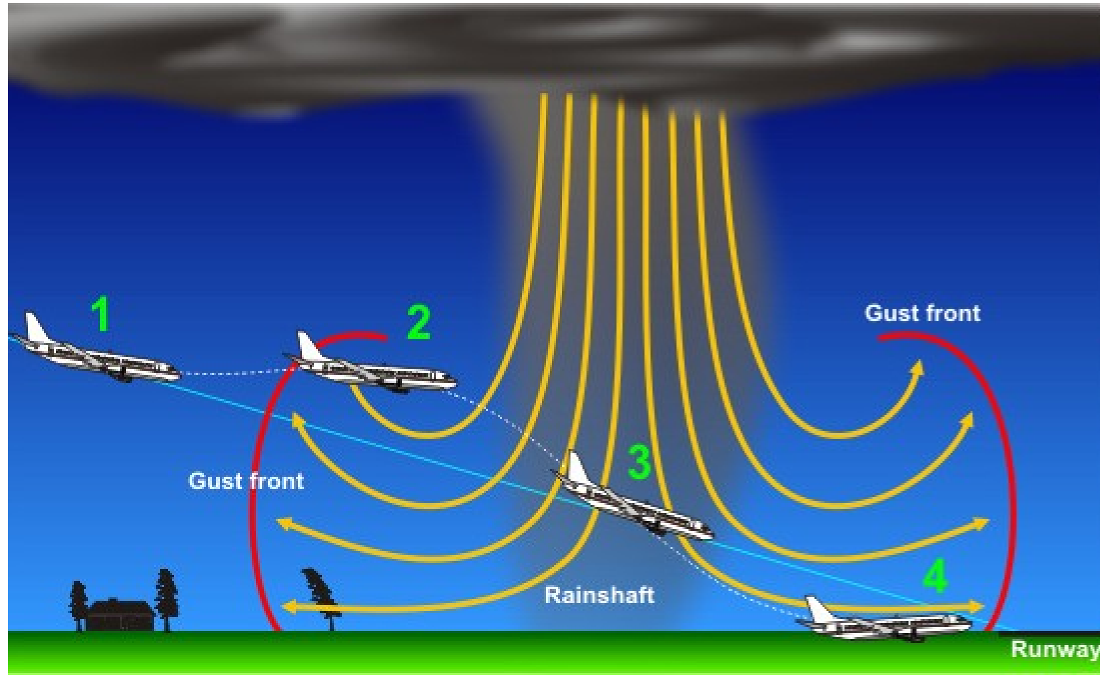
Daily facts



Cold drop situations and severe weather associated

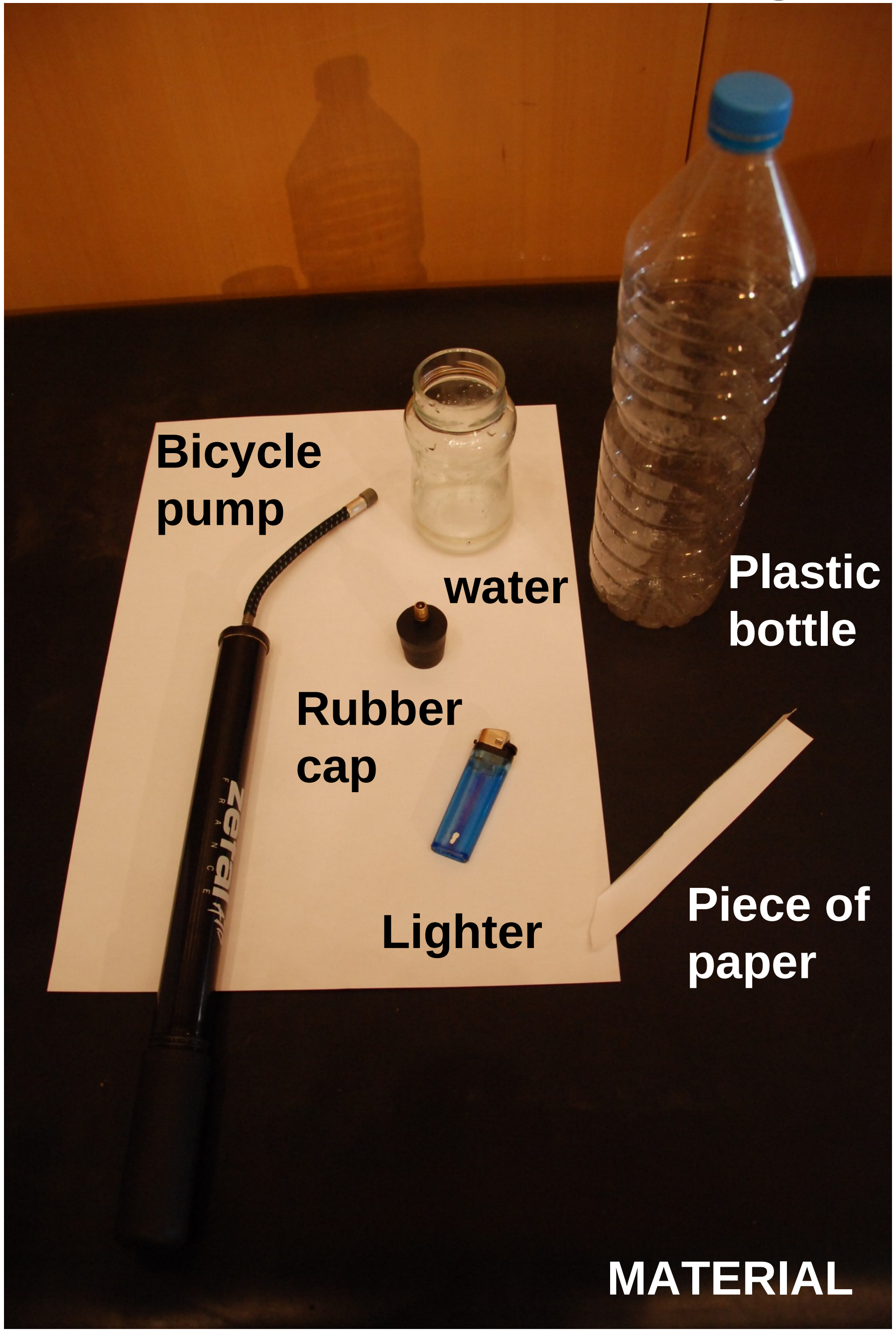
Daily facts





Downburst from cumulonimbus clouds and its effects

Let’s generate a cloud



1st ingredient:
Water vapor

2nd ingredient:
Condensation
nuclei (smoke)

Pressure
increase:
Air warming

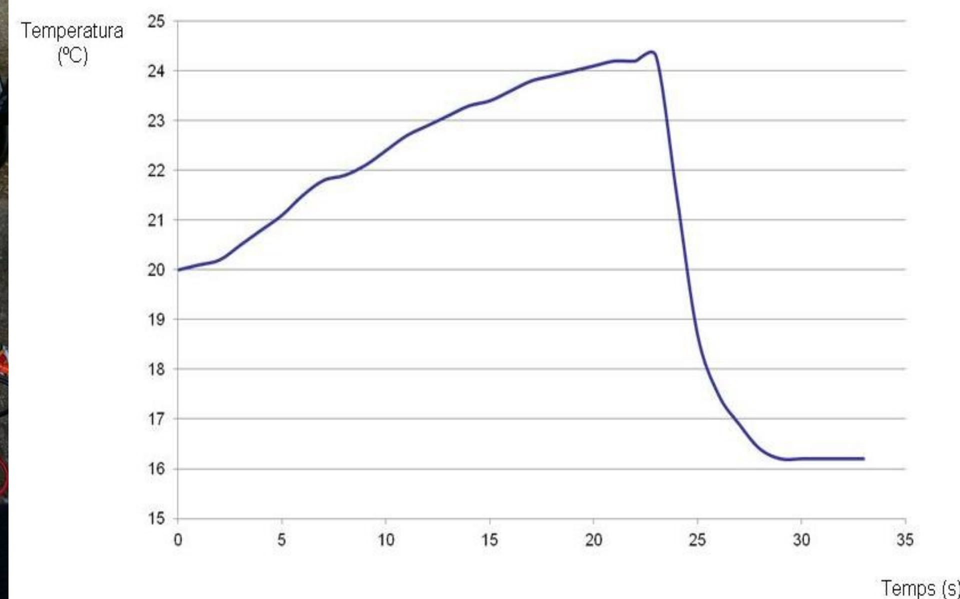
Let's generate a cloud

3rd ingredient:
Pressure decrease:
Air cooling

<u>Key concepts</u>: water vapor, air pressure, temperature, condensation nuclei.

<u>Key questions</u>: What happens when the pressure of moist air decreases?

Let’s generate a cloud



Installing sensors in the bottle

Air cooling in the adiabatic expansion

Daily facts

Skies and clouds

Pumping air